Visualizing US Natural Disaster Declarations: Trends and Patterns

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1. Introduction

Natural disasters such as floods, storms, wildfires, and hurricanes cause significant damage to human life and property. Analyzing disaster data can help understand patterns and trends over time. This project focuses on visualizing disaster declarations in the United States and identifying temporal trends using data visualization techniques. By analyzing yearly trends, disaster types, and seasonal patterns, we can gain useful insights into how disasters occur and change over time.

2. Project Description

The main objective of this project is to analyze and visualize natural disaster declarations in the United States. The project focuses on understanding how disasters change over time and identifying patterns in their occurrence.

The analysis includes:

- Studying the yearly trend of disaster declarations
- Comparing different types of disasters over time
- Identifying seasonal patterns in disasters across different months

Data visualization tools such as line charts and bar charts are used to represent the trends clearly. These visualizations help in understanding disaster patterns and possible environmental influences.

3. Dataset Used

The dataset used in this project is the **Disaster Declarations Summary dataset** provided by the **Federal Emergency Management Agency (FEMA)**.

This dataset contains historical records of disaster declarations in the United States. It includes information such as:

- Disaster declaration date
- Incident type (flood, storm, wildfire, etc.)
- State and location
- Incident start and end dates

For this analysis, the important columns used were:

- **declarationDate** – the date when the disaster was declared
- **incidentType** – the type of disaster
- **year** – extracted from the declaration date
- **month** – extracted from the declaration date

These fields helped analyze temporal patterns in disasters.

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4. Environment Setup

The analysis was performed using Python in a Jupyter Notebook environment.

The following tools and libraries were used:

- **Python** – programming language used for data analysis

- **Jupyter Notebook** – environment for writing and executing code
- **Pandas** – used for data manipulation and analysis
- **Matplotlib** – used for creating visualizations
- **Seaborn** – used for advanced data visualization

The dataset was loaded into the Jupyter Notebook using Pandas, and visualizations were generated using Matplotlib and Seaborn.

5. Data Exploration and Preprocessing

Before performing the analysis, the dataset was explored and cleaned to ensure accurate results.

The following preprocessing steps were performed:

1. **Loading the dataset**
   The CSV dataset was loaded into the Jupyter Notebook using the Pandas library.
2. **Checking dataset structure**
   The dataset structure was examined using functions like head() and info() to understand column names and data types.
3. **Converting date column**
   The declarationDate column was converted into datetime format to allow time-based analysis.
4. **Extracting time components**
   From the declaration date, the **year** and **month** columns were extracted for temporal analysis.
5. **Handling missing values**
   The dataset was checked for missing or invalid dates, and any missing values were removed to maintain data consistency.

6. After completing these preprocessing steps, the dataset was ready for visualization and trend analysis.

6. Yearly Trend Analysis – Insight

What does the graph show?
The graph shows how the number of disaster declarations in the United States changes over different years.

Pattern observed:
From the graph, we can see that the number of disasters has generally increased over the years. Some years show sudden increases compared to previous years.

Possible reason:
This increase may be because of climate change, better reporting systems, or the growing frequency of extreme weather events.

7.Incident Type Trends – Insight

What does the graph show?
This graph compares different types of disasters such as floods, storms, hurricanes, and other incidents across the years.

Pattern observed:
Storms and floods appear more frequently compared to other disaster types. Some disaster types show an increasing trend over time.

Possible reason:
Environmental changes and rising global temperatures may contribute to more storms and flooding events.

8.Monthly Seasonality – Insight

What does the graph show?
The graph shows the number of disaster declarations occurring in each month of the year.

Pattern observed:
Some months have a higher number of disasters than others. For example, months like March and September show higher disaster counts.

Possible reason:
This could be related to seasonal weather patterns such as hurricane seasons, heavy rainfall, or winter storms that occur during specific months.

9. Conclusion

In this project, disaster declaration data from the United States was analyzed to understand temporal trends and seasonal patterns.

The analysis revealed that disaster declarations have generally increased over time. Certain disaster types such as storms and floods occur more frequently compared to others. Additionally, the monthly analysis showed that disasters tend to occur more often in specific months, indicating seasonal patterns.

These insights highlight the importance of monitoring disaster trends and improving preparedness for natural disasters. Data visualization helps in better understanding these patterns and supports decision-making for disaster management.